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Multi-Agentive Generative AI Framework for Accelerating Field Development Planning

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Abstract

This paper introduces a multi-agentive solution that leverages Generative AI—specifically, Large Language Models (LLMs) coupled with domain-specific engines—to enhance the efficiency, consistency, and technical depth of reservoir simulation workflows. The solution targets three high-value areas: simulation model compliance, insight generation, and well placement optimization, with the goal of accelerating field development planning and institutionalizing engineering best practices.

The system is built around multiple AI agents, each integrating LLM-based natural language interfaces with specialized domain engines tailored for reservoir engineering tasks. The Reservoir Model Assessment Agent automates audits of simulation models by validating inputs, well constraints, and history matches internal modeling standards. The Reservoir Model Insights and Assessment Agent enables engineers to analyze, extract, and visualize critical model behaviors—such as production trends, scenario comparisons, and pressure evolution—through conversational queries. The Well Placement Optimization Agent blends simulation outputs with geospatial and operational constraints to generate ranked infill opportunities, simulate their performance impact, and provide rationale for each recommendation. All agents operate within a secure enterprise platform and are accessed via a unified, web-based interface that abstracts technical complexity while preserving engineering rigor.

Applied to a synthetic reservoir with over 100 wells sourced from various publicly available sources and multi-decade production history, the multi-agentive system demonstrated substantial performance improvements. Model compliance reviews were completed in less than 20% of the time required by traditional manual methods, with the AI agent identifying more than 85% of known deviations. Insight extraction tasks that typically took 2–3 days were reduced to under an hour, while enabling deeper analyses such as recovery factor diagnostics and scenario benchmarking. The well placement agent rapidly evaluated over 25 new infill candidates, identifying zones with up to 6.5% projected ultimate recovery uplift. The domain-specific engines embedded in each agent ensured high fidelity in technical outputs, while the natural language interface enabled broader accessibility for both junior and experienced engineers. The deployment also standardized workflows across teams, reduced reliance on tacit knowledge, and improved transparency in decision-making.

This work presents one of the first multi-agentic Generative AI solutions in reservoir engineering, combining the flexibility of LLMs with structured domain engines to deliver intelligent, explainable support across key simulation workflows. The result is a scalable, engineer-centric system that bridges AI and subsurface science to accelerate FDP and improve technical quality.

Introduction

Reservoir simulation is the backbone of field development planning (FDP), enabling engineers to analyze production scenarios, forecast reservoir performance, and support investment decisions. Despite decades of advancement in simulation tools and techniques, the practical execution of simulation workflows remains complex and time-consuming. Key activities—such as model validation, insight extraction, and well placement evaluation—are often manual, heuristic-driven, and reliant on domain expertise that is difficult to scale across teams or replicate consistently.

Recent progress in artificial intelligence (AI), particularly the emergence of Generative AI and Large Language Models (LLMs), offers new pathways for transforming engineering workflows. In broader technical domains, LLMs have demonstrated the ability to interpret unstructured inputs, generate structured outputs, and facilitate natural language interaction with complex systems. However, the adoption of these capabilities in reservoir engineering, and more specifically in reservoir simulation workflows, remains limited.

Early applications of LLMs in the upstream have focused on assistant-style tools designed to facilitate reservoir simulation setup and documentation. Similarly, [Wiegand et al. \(2024\)](#) proposed a generative AI assistant that helps engineers interact with simulation software through conversational interfaces to facilitate simulation setup tasks—such as preparing model input files, validating formats, or assisting with documentation. While promising, these systems primarily function as productivity aids, offering task-level acceleration without deeply engaging in technical reasoning, scenario diagnostics, or optimization.

In contrast, the framework presented in this paper introduces a multi-agentic generative AI system that embeds LLM capabilities within specialized, domain-aware agents, each purpose-built for core simulation tasks. Rather than focusing solely on setup or documentation, the proposed agents are designed to:

- Review reservoir model quality and compliance using internal modeling and simulation best practices
- Extract and contextualize insights from simulation outputs through natural language queries
- Optimize infill well placement with geospatial and performance ranking logic

Each agent integrates an LLM interface with a structured domain engine, ensuring technical fidelity and explainability, while enabling natural interaction across experience levels. This approach goes beyond assistant-style support and delivers an engineer-centric AI that directly addresses simulation challenges with smart, explainable automation.

The remainder of the paper outlines the agent methodologies, agent capabilities, and results from a synthetic field case with over 100 wells. The work represents one of the first multi-agent generative AI applications in reservoir simulation and contributes to the broader digital transformation efforts underway in upstream oil and gas.

Methodology

Multi-Agentic Framework Overview

The methodology employs a multi-agentic framework that utilizes Generative AI, specifically LLMs to address field development planning tasks in reservoir engineering. The core idea is to create a system

composed of distinct AI agents, each tailored to specific tasks within the workflow, thereby enhancing operational efficiency and technical precision. The framework is structured around independent agents, each of which leverages the capabilities of LLMs as its reasoning engine while integrating domain-specific tools to support specialized workflows. These agents work collaboratively, with the ability to communicate and share knowledge, ensuring a comprehensive and dynamic approach to problem-solving.

The agents are goal-driven systems that interact with domain-specific engines and external tools to autonomously execute tasks, trigger workflows, and connect with simulation functions. They are managed by an LLM router, which directs the agents to the appropriate resources, ensures alignment with their goals, and coordinates their activities to achieve the best outcomes.

To facilitate interaction between the LLM agents and domain-specific tools, the system is structured as follows and as shown in Fig 1:

1. **Agent Core:** Each agent is built around a central processing unit, the Agent Core, which interprets the input from users, applies reasoning to determine the next steps, and decides on the appropriate actions. This core integrates the LLM's reasoning capabilities with specialized engines to execute tasks with high fidelity.
2. **Tools and Workflows:** The system is equipped with a variety of external tools that extend the agent's capabilities. These tools include domain-specific engines (e.g. reservoir simulator, well placement, waterflood optimizer) that allow the agents to perform specialized tasks like running simulations, placing wells, optimizing waterflood, and more. The tools enable agents to execute predefined workflows, such as hotspot identification, placing infill wells, or running reservoir simulation based on the user's inputs.
3. **Planning Module:** A key element of the system is the Planning module, which assists the agents in problem-solving by using techniques such as the Chain-of-Thought approach to break down complex tasks into smaller, manageable steps. This approach ensures that even intricate workflows are executed in a structured and logical manner, optimizing the decision-making process.
4. **Memory:** Both short-term and long-term memory are critical for task execution. Short-term memory helps agents retain immediate context to ensure coherent task execution, while long-term memory stores historical data, previous decisions, and learned insights to support ongoing tasks and improve system performance over time.
5. **LLM Router:** The LLM Router plays a crucial role in managing interaction between multiple agents, ensuring that each agent is directed to the relevant tools, data, and workflows based on the task at hand. This system ensures that each agent remains focused on its specific goal, while also facilitating the exchange of knowledge and data between agents when necessary.

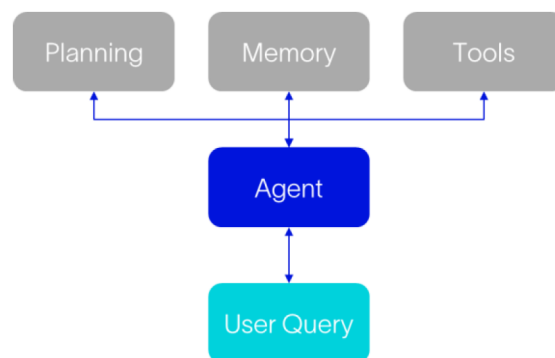


Figure 1—LLM Agent Architecture

Integration of LLMs with Domain-Specific Engines

The integration of LLMs with domain-specific engines forms the foundation of the multi-agentic system, combining the reasoning capabilities of LLMs with the specialized functionality of reservoir engineering tools. This approach allows the system to handle complex, multi-dimensional tasks that would be difficult for a single agent or model to tackle alone. LLMs provide the ability to interpret and generate natural language-based inputs and outputs, while the domain-specific engines deliver the technical depth and precision required for tasks like reservoir simulation, optimization, and scenario analysis.

Each agent in this multi-agentic system is designed to achieve specific goals within the field development planning (FDP) process:

1. **Reservoir Model Assessment Agent** — executes compliance reviews against internal modeling / simulation standards.
2. **Reservoir Insights & Assessment Agent** — answers analytical questions and produces plots/tables for trends, scenario comparisons, and pressure evolution.
3. **Well-Placement Optimization Agent** — generates and ranks infill opportunities under geospatial and operational constraints and simulates expected performance impact.

The architecture presented in Fig 2 depicts the multi-agentic workflow that was implemented:

- **Step 1:** The user's query is first parsed by the LLM router agent that identifies intent and decomposes the task. Based on the detected intent, the router dispatches one or more domain agents in parallel.
- **Step 2:** Each agent can call specific domain functions (e.g., simulators, engine scripts, databases, APIs) and retrieve data from manuals/guidelines and model data.
- **Step 3:** Outputs from the agents, such as findings, retrieved snippets, numeric results, and constraints, are consolidated into a structured prompt.
- **Step 4:** The LLM consumes the updated prompt, reasons over the agent outputs, and synthesizes a response. A Memory store maintains dialog context, enabling continuity across turns and improving subsequent routing and retrieval.
- **Step 5:** The system returns a concise answer and recommends next actions.

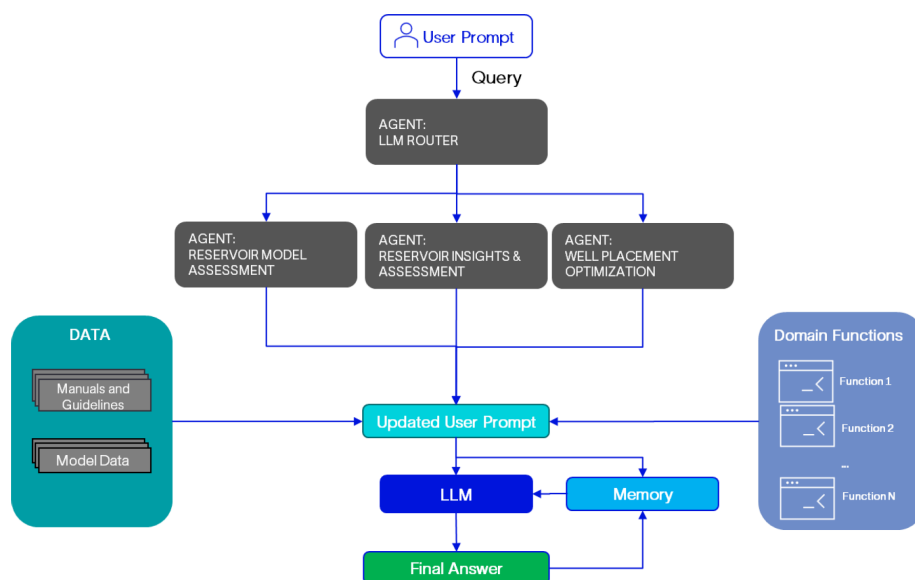


Figure 2—Multi-Agentic Framework Architecture

The modular layout allows additional agents/functions to be added without changing the orchestration layer. The agents operate within a secure enterprise platform that abstracts much of the technical complexity, providing users with an intuitive interface that simplifies interaction while maintaining rigorous engineering standards. The platform facilitates the execution of the full simulation lifecycle, from model parsing and compliance checks to insight generation and well placement optimization. By connecting seamlessly with existing simulation models and external domain tools, the multi-agent system provides a robust, scalable solution for field development planning.

AI Agents

Reservoir Model Assessment Agent

The Reservoir Model Assessment Agent automates the validation of reservoir simulation models against operator's internal standards, corporate modeling guidelines, and established best practices, significantly reducing the time and effort required for manual model reviews. By combining an LLM-powered conversational interface with a domain-specific compliance engine, the agent systematically reviews simulation inputs, configuration files, and operational parameters, producing clear, actionable assessments of model quality and conformance.

The agent connects simulation input files—including 3D grid properties, well connections and trajectories, relative permeability and capillary pressure tables, PVT datasets, vertical flow performance (VFP) tables, field management logic—and simulation output files such as time-series production and injection results, and time-series 3D grid property evolution (oil saturation, reservoir pressure, gas saturation, water cut), along with operator specific modeling and simulation guidelines, internal best practices, reservoir management guidelines, and more. This enables a comprehensive compliance check that covers both model configuration and performance alignment with historical data. Engineers can query the agent in natural language to pinpoint non-conformities, understand their potential impact, and receive targeted recommendations for corrective action.

The following are examples of core functionality:

- **Input Parameter Compliance Review** – Verifies that static and dynamic reservoir properties, relative permeability and capillary pressure data, PVT and VFP tables, and other key input parameters conform to operator standards and reservoir management guidelines.
- **Well and Operational Constraints Validation** – Checks that well controls, production/injection limits, and operational parameters align with prescribed corporate constraints and field development policies.
- **History Match Quality Assessment** – Evaluates the fidelity of the history match against defined quality ranges, identifying mismatches in production, injection, and pressure profiles, and flagging areas requiring recalibration.
- **Compliance Reporting and Recommendations** – Generates structured compliance summaries highlighting deviations, their severity, and suggested corrective measures, which can be exported for audit records or integrated into model update workflows.

By automating comprehensive model reviews, the Reservoir Model Assessment Agent compresses multi-day review processes into hours, ensures consistent application of corporate standards, and improves the technical integrity of simulation models before they are used for scenario evaluation or optimization.

Reservoir Insights & Assessment Agent

The Reservoir Model Insights & Assessment Agent provides engineers with direct, structured access to the complete spectrum of simulation input datasets and output results, enabling comprehensive interrogation of reservoir models through natural language queries. By integrating an LLM-powered conversational

interface with a domain-specific engine connected to simulator input and output data, the agent unifies static and dynamic reservoir information into a single interactive workflow. This allows engineers to seamlessly transition from reviewing model setup to diagnosing in-situ reservoir behavior without switching tools or manually extracting data.

The following are examples of core functionality:

- **Simulation Input Data Analysis** – Provides a rapid, structured overview of the simulation model by retrieving and synthesizing all static input datasets, including 3D grid properties, well connections and trajectories, relative permeability and capillary pressure tables, PVT datasets, VFP tables, and field management logic. This functionality enables engineers to quickly understand model architecture, reservoir characterization, and operational constraints without manually navigating complex input files.
- **Dynamic Reservoir Dynamics Analysis** – Interprets time-evolving reservoir states by visualizing the spatial and temporal progression of key 3D properties such as oil and gas saturation fronts, pressure propagation, and coning behavior. Automated insights highlight significant performance drivers, anomalous patterns, and emerging risks, enabling faster identification of reservoir management opportunities.
- **Reservoir, Group, and Well Performance Diagnostics** – Evaluates performance across the entire production by analyzing historical and/or forecasted production and injection trends, water cut and gas-oil ratio progression, pressure support effectiveness, and variances between simulated and observed behavior. This functionality supports both macro-level field diagnostics and micro-level well troubleshooting.
- **Scenario Benchmarking** – Performs side-by-side comparisons of multiple development strategies, quantifying differences in recovery factors, production plateau durations, breakthrough timing, sweep patterns, and regional performance distribution. This enables objective ranking of scenarios, identification of the most impactful design changes, and validation of optimization strategies.

Through its conversational interface, the agent blends textual and visual outputs—maps, cross-sections, trend plots, and KPI tables—directly in response to user queries. Domain-specific prompt engineering ensures that the LLM outputs remain technically accurate, contextually relevant, and reproducible, while the underlying domain engine executes complex queries and numerical computations against the simulation database.

By enabling immediate, integrated access to both static and dynamic datasets, the Reservoir Model Insights & Assessment Agent compresses multi-day analytical workflows into minutes, increases the depth and consistency of evaluations, and provides engineers with a scalable, repeatable framework for diagnosing model behavior and optimizing field development strategies.

Well Placement Optimization Agent

The Well Placement Optimization Agent is designed to identify, evaluate, and recommend optimal locations for new infill wells by integrating reservoir simulation outputs, geospatial constraints, and operational considerations into a single, AI-driven workflow. It combines a domain-specific optimization engine with an LLM-powered natural language interface, enabling engineers to execute complex placement studies interactively, while ensuring consistency, explainability, and engineering rigor.

The methodology is implemented as an end-to-end automated pipeline consisting of two key stages:

Stage 1: Base Case Field Development Scenario Analysis. In the first stage, the agent analyzes the current (base) FDP simulation case to establish performance baselines and identify underperforming regions:

- **Well Statistics & Insights** – Automatically categorizes drilled wells by type, development region, and drilling schedule, producing a spatial and temporal profile of existing field development.
- **Performance Diagnostics** – Evaluates regional performance using recovery factors, production/injection trends, water cut, gas–oil ratio, and pressure maintenance metrics.
- **Opportunity Screening** – Ranks development regions and wells by performance gaps, highlighting underperforming infill candidates and identifying zones where well replacement or new drilling may yield the highest incremental recovery.

This stage leverages fully scripted data handling and analytics routines, ensuring rapid turnaround and consistent evaluation criteria across multiple simulation scenarios. User can potentially change the sequence of steps, introduce new ones or remove any of the existing.

Stage 2: Automated and AI-Enabled Well Placement Workflow. The second stage applies to the agent's optimization capabilities to generate, simulate, and evaluate new well placement options:

- **Opportunity Index Mapping** – Integrates 3D static and dynamic properties to quantify and visualize high-potential drilling zones, with probabilistic thresholds in case of multiple-scenario assessments.
- **Property Combination & Risk Weighting** – Combines opportunity indices with additional metrics (e.g., drilling risk, completion quality) using configurable methods such as weighted sums or root-product combinations.
- **Existing Well Proximity Filtering** – Enforces geospatial constraints to prevent interference with existing wells, completions and surface infrastructure.
- **Well Placement & Trajectory Design** – Automatically generates well trajectories for different well types including and not limited to well patterns (e.g., 5-spot, peripheral, line-drive) with user-defined limits on spacing, length, azimuth, and segment count, while honoring reservoir boundaries and no-go zones.
- **Completions & Facility Placement** – Designs completion intervals based on reservoir properties, integrates facility location planning (e.g., pad or platform slots), and incorporates these constraints into the optimization process.
- **Simulation Integration & Evaluation** – Embeds generated wells into the reservoir simulation model, launches automated runs, and assesses production forecasts—both deterministic and probabilistic—against defined objective functions.

The steps described here can be called and executed separately or together as a single pipeline. Through automation and embedded AI logic, the Well Placement Optimization Agent reduces the cycle time for infill planning from weeks to hours, while enabling engineers to explore more scenarios and maintain rigorous technical standards. The LLM interface allows engineers to query results conversationally and receive outputs.

This agent's design ensures that opportunity identification, well design, and simulation validation are tightly coupled in a repeatable workflow—bridging subsurface insight, operational feasibility, and reservoir performance optimization.

Human-in-the-Loop Example of Agent Capabilities

This section illustrates an end-to-end workflow where multiple AI agents collaborate together with the engineer to optimize a field development plan. The scenario demonstrates how the Reservoir Model Assessment, Reservoir Insights & Assessment Agent, Simulator Syntax Generator Agent, and Well Placement Optimization Agent work together to identify underperforming wells, modify the simulation model, and propose new well placements – all within a single, unified interface.

Use Case: Infill Drilling Optimization. The objective of this exercise is to analyze an existing baseline FDP reservoir simulation model, identify underperforming future wells, and replace them with better-performing alternatives. This workflow showcases how Generative AI can streamline traditionally manual tasks and enhance decision-making through Agentic workflows keeping the engineer always in the loop.

Step 1: Identifying Underperforming Wells

The engineer begins by querying the model to list wells drilled before, after or in a specific year (e.g., 2000). The system returns a list of wells. Afterwards, user asks couple more follow up questions and gets list of wells with lowest cumulative production and production profiles. Wells with the lowest production—such as wells #1 and #2—are flagged for potential replacement.

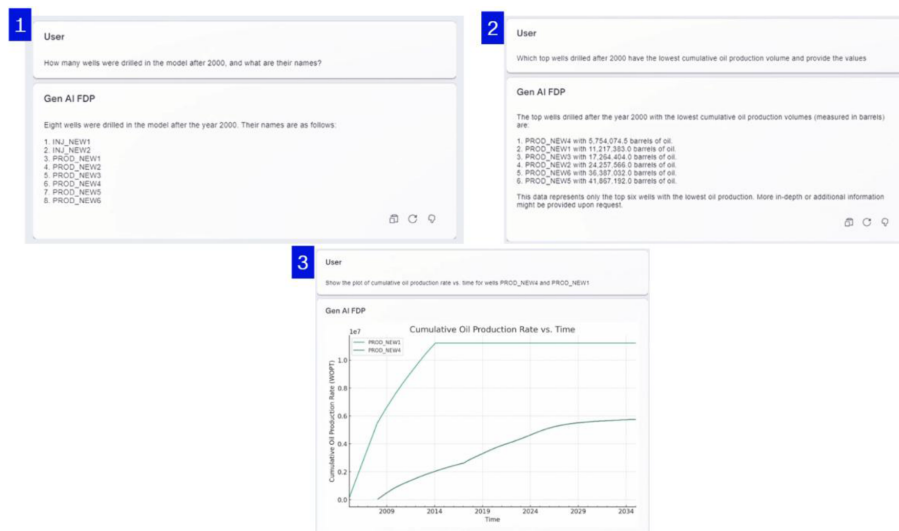


Figure 3—Step 1 user queries and results

Step 2: Visualizing Well Locations

Next, the engineer requests a 2D visualization of the reservoir model showing the spatial location of the wells. The map overlays properties such as opportunity index or porosity, helping the engineer contextualize well performance within the reservoir.

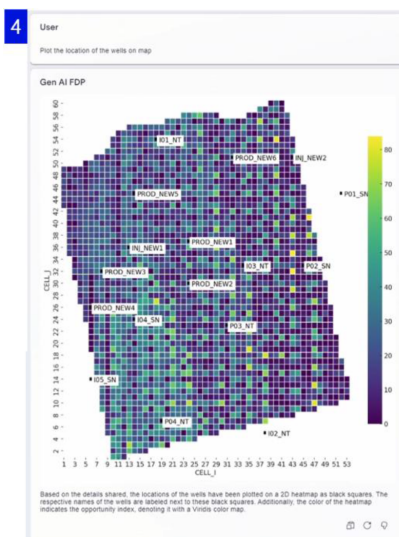


Figure 4—Step 2 user queries and results

Step 3: Modifying the Simulation Model

To assess the impact of removing underperforming wells, the engineer instructs the system to close wells #1 and #4. The Simulator Syntax Generator Agent automatically generates the required simulation syntax and updates the simulation deck, eliminating the need for manual edits.

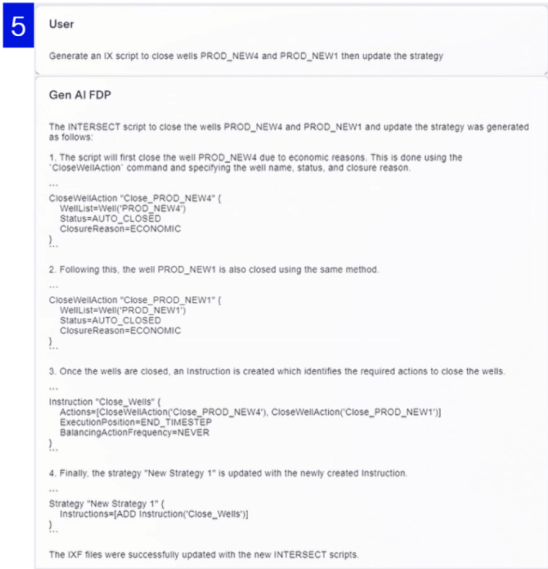


Figure 5—Step 3 user queries and results

Step 4: Calculating Opportunity Index

The engineer then requests the calculation of the opportunity index to identify optimal zones for new well placement. The system uses either industry-standard or custom-defined formulas to generate a spatial opportunity map.

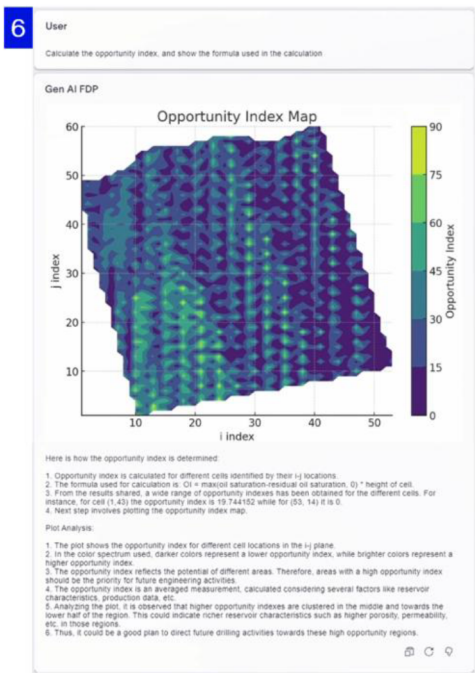


Figure 6—Step 4 user queries and results

Step 5: Automated Well Placement

Using the opportunity index and user-defined constraints (e.g., number of producers/injectors, spacing, cost), the Well Placement Optimization Agent generates a new well placement scenario. It designs well trajectories, integrates completions, and prepares simulation-ready inputs.

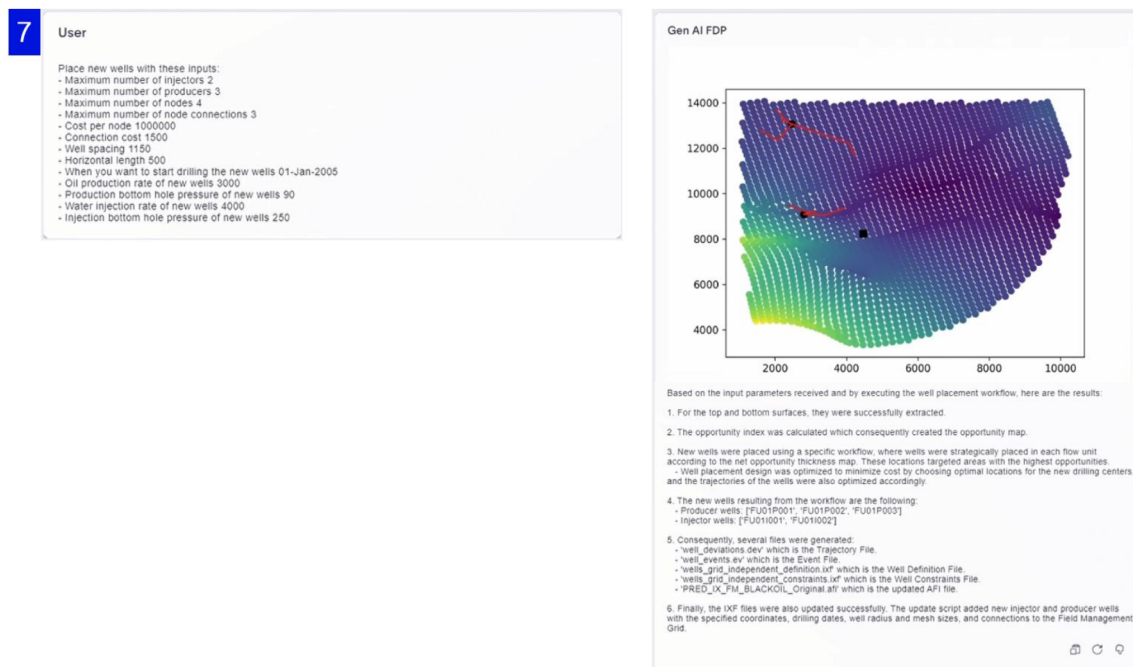


Figure 7—Step 5 user queries and results

Step 6: Simulation and Performance Comparison

The updated model is simulated, and the engineer queries the system to compare production results. The newly placed wells (#1, #2) show significantly higher cumulative production (32M and 14M barrels) compared to the removed wells (5M and 11M barrels), confirming the success of the optimization.

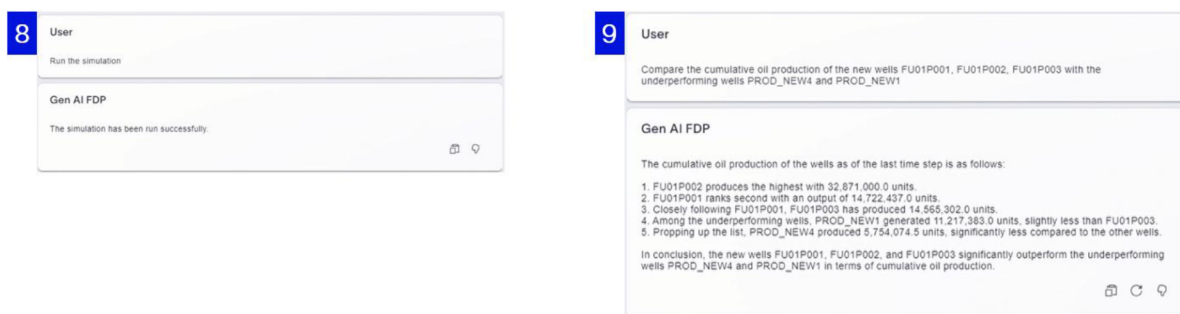


Figure 8—Step 6 user queries and results

Case Study

To evaluate the applicability of the multi-agentic Generative AI framework in a realistic setting, the system was tested on a synthetic reservoir case representing a mature field with more than 100 production and injection wells and over 25 years of production history. The dataset incorporated heterogeneous facies distribution, strong aquifer support, and multiple pressure-depletion phases, making it representative of real-world complexity.

The workflow focused on three primary objectives: (1) ensuring model compliance and readiness for decision-making, (2) generating actionable insights for reservoir management, and (3) identifying infill opportunities through automated well placement.

Model Compliance and Validation

The Reservoir Model Assessment Agent was applied to the baseline simulation deck containing over 400,000 active cells. The agent identified 15 deviations relative to internal modeling standards, including inconsistent PVT correlations, and well constraints outside operator's reservoir management guidelines such as bottom hole pressure or gas-oil ratio constraints. Corrective recommendations were automatically generated, enabling engineers to address the issues in less than a minute—compared to the several hours typically required for manual audits.

Reservoir Insights and Diagnostics

The Reservoir Insights & Assessment Agent was queried to examine water breakthrough patterns, recovery factor evolution under waterflood for each development unit, and cross-scenario comparisons. The system produced a structured summary highlighting that injector dynamics leveraging streamlines data in the southern flank contributed to early water breakthroughs in three producers. These insights were generated in less than few minutes, replacing what normally required hours to days of manual extraction, plotting, and interpretation for large number of wells.

Well Placement Optimization

The Well Placement Optimization Agent was then applied to identify new infill opportunities in under-swept regions. Using a combination of opportunity index map and geospatial constraints, the system generated 50 candidate wells. After automated filtering for spacing and operational feasibility, 25 candidates were selected for simulation. Comparative performance runs demonstrated that 19 wells consistently yielded incremental recovery gains, with the best-case offering a 6.5% uplift in ultimate recovery relative to the base case together with pressure support scenario. Engineers could interactively query the rationale behind each recommendation, ensuring transparency in the optimization logic.

Overall, the case study demonstrated that the multi-agent framework not only reduced cycle times but also enabled deeper, more consistent analysis across multiple decision points.

Conclusion

This work presents one of the first documented applications of a multi-agentic Generative AI framework in reservoir simulation and field development planning. By integrating LLM-based reasoning with domain-specific engines, the system enabled:

- Automation of compliance reviews, ensuring models meet internal standards in a fraction of the time traditionally required.
- Acceleration of insight generation, allowing engineers to rapidly diagnose reservoir behavior and benchmark scenarios through conversational interaction.
- Optimization of well placement, providing a systematic and explainable approach to infill drilling recommendations.

The case study highlights that such a framework can reduce manual effort, increase consistency across teams, and uncover non-trivial optimization opportunities that might otherwise be missed under time constraints. Importantly, the natural language interface broadened accessibility, allowing both junior and senior engineers to interrogate models and contribute to decision-making with minimal learning curve.

While the current implementation was demonstrated on a synthetic dataset, the architecture is directly extensible to real-field applications, with potential to integrate additional agents such as waterflood

optimization, gas lift optimization, economic evaluation, uncertainty quantification, and more. Future work will focus on expanding the scope of domain-specific engines covering full-scale field development planning activities, incorporating reinforcement learning for adaptive optimization, and testing deployment on live assets.

The findings demonstrate that multi-agentic Generative AI has the potential to fundamentally transform reservoir simulation workflows—moving from manual, expert-driven tasks toward intelligent, explainable, and collaborative AI systems that accelerate field development planning and improve technical quality.

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